4-Tensors

Contravariant 4-vector $x^\alpha = (ct, \vec{r}) = (ct, x, y, z)$ $\alpha = 0, 1, 2, 3$

We found
$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$
transformation from K to K'.

Write it as $x'^\alpha = \sum_{\beta=0}^{3} L^\alpha{}_\beta \, x^\beta = L^\alpha{}_\beta \, x^\beta$ (Einstein convention)

$$\Rightarrow L^\alpha{}_\beta = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

General contravariant vector = set of 4 quantities A^α which transform as x^α: $A'^\alpha = L^\alpha{}_\beta A^\beta$

Contravariant 4-tensor $T^{\alpha\beta\gamma\ldots}$ is object that transforms according to $\boxed{T'^{\alpha\beta\gamma\ldots} = L^\alpha{}_\varkappa L^\beta{}_\mu L^\gamma{}_\nu T^{\varkappa\mu\nu\ldots}}$

In other words, $T^{\alpha\beta\gamma}$ <u>transforms as product</u> of three 4-vectors $T^{\alpha\beta\gamma} = A^\alpha B^\beta C^\gamma$

(since $A'^\alpha = L^\alpha{}_\varkappa A^\varkappa$, $B'^\beta = L^\beta{}_\mu B^\mu$, $C'^\gamma = L^\gamma{}_\nu C^\nu$).

Importantly, $T^{\alpha\beta\gamma} \neq A^\alpha B^\beta C^\gamma$. But it transforms the same way such a product would.

Note that from $x'^\alpha = L^\alpha{}_\beta x^\beta$ it follows that
$$dx'^\alpha = L^\alpha{}_\beta \, dx^\beta$$
(we used this deriving how velocity transforms)

But $dx'^\alpha = \dfrac{\partial x'^\alpha}{\partial x^\beta} dx^\beta \Rightarrow L^\alpha{}_\beta = \dfrac{\partial x'^\alpha}{\partial x^\beta}$.

Hence $A'^\alpha = \dfrac{\partial x'^\alpha}{\partial x^\beta} A^\beta$.

Covariant 4-vector $X'_\beta = M^\alpha{}_\beta X_\alpha$

But $X'_\beta = (ct', -\vec{r}\,') \to (ct', -x')$

We know $\begin{pmatrix} ct' \\ x' \end{pmatrix} = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}$

but $\begin{pmatrix} ct' \\ -x' \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} ct' \\ x' \end{pmatrix}$

and $\begin{pmatrix} ct \\ x \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} ct \\ -x \end{pmatrix}$

$$\Rightarrow \begin{pmatrix} ct' \\ -x' \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \underbrace{\gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}}_{\gamma \begin{pmatrix} 1 & \beta \\ -\beta & -1 \end{pmatrix}} \begin{pmatrix} ct \\ -x \end{pmatrix} = \gamma \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & \beta \\ -\beta & -1 \end{pmatrix} \begin{pmatrix} ct \\ -x \end{pmatrix} =$$

$$= \gamma \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix} \begin{pmatrix} ct \\ -x \end{pmatrix}$$

$$\Rightarrow \text{matrix } M^\alpha{}_\beta = (L^{-1})^\alpha{}_\beta = \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$X'_\beta = (L^{-1})^\alpha{}_\beta X_\alpha$$

Note that $(L^{-1})^\alpha{}_\beta L^\beta{}_\gamma = \delta^\alpha_\gamma = L^\alpha{}_\beta (L^{-1})^\beta{}_\gamma$

e.g. $\gamma \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix} \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} = \gamma^2 \begin{pmatrix} 1-\beta^2 & 0 \\ 0 & 1-\beta^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$

Covariant 4-tensor $T_{\alpha\beta\ldots}$ is an object that <u>transforms</u> as

$$T'_{\alpha\beta\ldots} = (L^{-1})^\varkappa{}_\alpha (L^{-1})^\mu{}_\beta \cdots T_{\varkappa\mu\ldots}$$

Again, $T_{\alpha\beta} \sim A_\alpha B_\beta$

Mixed tensor $T^{\alpha}{}_{\beta} \sim A^{\alpha} B_{\beta}$

$\Rightarrow$ it transforms as $\left(A'^{\alpha} = L^{\alpha}{}_{\beta} A^{\beta}, \quad B'_{\beta} = (L^{-1})^{\mu}{}_{\beta} B_{\mu} \right)$

$$T'^{\alpha}{}_{\beta} = L^{\alpha}{}_{\gamma} (L^{-1})^{\mu}{}_{\beta} T^{\gamma}{}_{\mu}$$

Note that generally $T^{\alpha}{}_{\beta} \neq T_{\beta}{}^{\alpha}$

since $T'_{\beta}{}^{\alpha} = (L^{-1})^{\mu}{}_{\beta} L_{\gamma}{}^{\alpha} T_{\mu}{}^{\gamma}$.

Few examples:

1) how T^{00} transforms?

T'^{00} transforms the same way as $A'^{0} B'^{0}$

$$A'^{0} = \gamma A^{0} - \gamma\beta A^{1}, \quad B'^{0} = \gamma (B^{0} - \beta B^{1})$$

$$\Rightarrow A'^{0} B'^{0} = \gamma^{2} (A^{0} - \beta A^{1})(B^{0} - \beta B^{1}) = \gamma^{2} (A^{0} B^{0} - \beta A^{0} B^{1} - \beta A^{1} B^{0} + \beta^{2} A^{1} B^{1})$$

$$\Rightarrow T'^{00} = \gamma^{2} (T^{00} - \beta T^{01} - \beta T^{10} + \beta^{2} T^{11})$$

For inverse transformation $\beta \to -\beta$.

2) how $T^{0}{}_{0}$ transforms?

Now we need to think of $A^{0} B_{0}$

But $B'_{0} = \gamma (B_{0} + \beta B_{1})$ $\leftarrow$ note the sign change!

$$\Rightarrow A'^{0} B'_{0} = \gamma^{2} (A^{0} - \beta A^{1})(B_{0} + \beta B_{1}) = \gamma^{2} (A^{0} B_{0} + \beta A^{0} B_{1} - \beta A^{1} B_{0} - \beta^{2} A^{1} B_{1})$$

$$\Rightarrow T'^{0}{}_{0} = \gamma^{2} (T^{0}{}_{0} + \beta T^{0}{}_{1} - \beta T^{1}{}_{0} - \beta^{2} T^{1}{}_{1})$$

We know that the interval $s^2 = x_\alpha x^\alpha = x^\alpha x_\alpha = (ct)^2 - (\vec{r})^2$ is Lorentz invariant.

It is a scalar (number) and does not change under transformation.

On the other hand, $s^2 = x'^\alpha x'_\alpha$

$$x'^\alpha = L^\alpha{}_\beta x^\beta \quad , \quad x'_\alpha = (L^{-1})^\gamma{}_\alpha x_\gamma$$

$$\Rightarrow \quad x'^\alpha x'_\alpha = L^\alpha{}_\beta x^\beta (L^{-1})^\gamma{}_\alpha x_\gamma = L^\alpha{}_\beta (L^{-1})^\gamma{}_\alpha x^\beta x_\gamma$$

$$= \underbrace{(L^{-1})^\gamma{}_\alpha L^\alpha{}_\beta}_{\delta^\gamma{}_\beta} x^\beta x_\gamma = \delta^\gamma{}_\beta x^\beta x_\gamma = x^\beta x_\beta .$$

Permutation of indices:

symmetric tensor $T^{\alpha\beta} = T^{\beta\alpha}$

antisymmetric $T^{\alpha\beta} = -T^{\beta\alpha}$

arbitrary $T^{\alpha\beta} = \underbrace{\frac{1}{2}(T^{\alpha\beta} + T^{\beta\alpha})}_{\text{symmetric}} + \underbrace{\frac{1}{2}(T^{\alpha\beta} - T^{\beta\alpha})}_{\text{antisymmetric}}$

<u>Quotient rule</u>: helps to determine if an entity $T^{\alpha\beta}{}_{\gamma\delta\ldots}$ is a tensor. If contraction of $T^{\alpha\beta}{}_{\gamma\delta\ldots}$ with <u>arbitrary</u> tensor $B_{\mu\nu}$ gives a tensor $A^{\alpha\beta\ldots}$, then $T^{\alpha\beta}{}_{\gamma\delta\ldots}$ is a tensor.

Consider for example $T^{\alpha\beta} B_\beta = A^\alpha$ (contraction always involves contravariant and covariant indices.)

do not know if this is a tensor

4-vector

In K': $T'^{\alpha\beta} B'_\beta = A'^\alpha$

$$T'^{\alpha\beta} (L^{-1})^\mu{}_\beta B_\mu = L^\alpha{}_\nu A^\nu = L^\alpha{}_\nu T^{\nu\lambda} B_\lambda$$

since B_μ is arbitrary

$$\Rightarrow T'^{\alpha\beta} (L^{-1})^\mu{}_\beta = L^\alpha{}_\nu T^{\nu\mu}$$

Multiply by $L^\lambda{}_\mu$: $T'^{\alpha\beta} \underbrace{(L^{-1})^\mu{}_\beta L^\lambda{}_\mu}_{L^\lambda{}_\mu (L^{-1})^\mu{}_\beta = \delta^\lambda{}_\beta} = T'^{\alpha\beta} \delta^\lambda{}_\beta = T'^{\alpha\lambda}$

$$\Rightarrow T'^{\alpha\lambda} = L^\alpha{}_\nu T^{\nu\mu} L^\lambda{}_\mu = L^\alpha{}_\nu L^\lambda{}_\mu T^{\nu\mu}$$

Therefore we see $T^{\alpha\lambda}$ is a tensor because it transforms as one!

— Consider 4-velocity $u^\alpha = \frac{dx^\alpha}{d\tau} = \gamma \frac{dx^\alpha}{dt} = \gamma \frac{d}{dt}(dct, d\vec{r}) = \gamma(c, \vec{v})$.

where $d\tau$ = proper time $= \frac{ds}{c}$

$d\tau = \frac{dt}{\gamma}$

Now, $dx^\alpha = u^\alpha d\tau$ and $d\tau$ = scalar

$\Rightarrow u^\alpha$ is 4-vector indeed.

— Phase of a wave $\phi = \omega t - \vec{k}\cdot\vec{r} = k_\alpha x^\alpha$

define $k_\alpha = \left(\frac{\omega}{c}, -\vec{k}\right)$

ϕ is invariant — all observers count the same number of wave fronts (although the wave length is not invariant)

$\Rightarrow x^\alpha$ is 4-vector, ϕ = scalar $\Rightarrow k_\alpha$ = 4-vector too.

$k^\alpha = \left(\frac{\omega}{c}, \vec{k}\right)$

Relativistic metric tensor g

$$\left.\begin{aligned} A_\alpha &= g_{\alpha\beta} A^\beta \\ A^\beta &= g^{\beta\gamma} A_\gamma \end{aligned}\right\} \quad A_\alpha = \underbrace{g_{\alpha\beta}\, g^{\beta\gamma}}_{\delta_\alpha{}^\gamma = \delta^\gamma{}_\alpha} A_\gamma$$

$\Rightarrow$ Metric tensor $g_{\alpha\beta}$ is its own inverse.

Use it to move indices: contravariant $\longleftrightarrow$ covariant.

In particular, $x_\alpha = g_{\alpha\beta} x^\beta$

But $s^2 = x_\alpha x^\alpha = g_{\alpha\beta} x^\beta x^\alpha = g_{\alpha\beta} x^\alpha x^\beta = c^2 dt^2 - x^2 - y^2 - z^2$

$$\Rightarrow g_{\alpha\beta} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = g^{\alpha\beta}$$ is the same in all coordinate systems,

e.g. $$K^\alpha = g^{\alpha\beta} K_\beta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \frac{\omega}{c} \\ -k_x \\ -k_y \\ -k_z \end{pmatrix} = \begin{pmatrix} \frac{\omega}{c} \\ +k_x \\ +k_y \\ +k_z \end{pmatrix} = \left(\frac{\omega}{c}, \vec{k}\right)$$

$$g'^{\alpha\beta} = L^\alpha{}_\mu L^\beta{}_\nu g^{\mu\nu} = g^{\alpha\beta}$$

$$\Rightarrow \det(g^{\alpha\beta}) = \det^2(L) \det(g)$$

$$\Rightarrow \det^2(L) = 1$$

$$\begin{cases} \text{Lorentz transformation} & \det(L) = 1 \\ \text{Reflections} & \det(L) = -1 \end{cases}$$